STOR 557 HW 7

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```
boxplot(coag ~ diet, data = coagulation,
        xlab = "Diet",
        ylab = "Coagulation Time",
        main = "Coagulation Time by Diet")
```



Diet C has the highest coagulation times, while diets A and D have the lowest. Diets B and D show more variability, and diet C has a possible outlier.

```
m1 <- lm(coag ~ diet, data = coagulation)

summary(m1)
```

```
##
## Call:
```

```
## lm(formula = coag ~ diet, data = coagulation)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -5.00  -1.25   0.00   1.25   5.00
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.100e+01  1.183e+00  51.554  < 2e-16 ***
## dietB       5.000e+00  1.528e+00   3.273 0.003803 **
## dietC       7.000e+00  1.528e+00   4.583 0.000181 ***
## dietD       2.719e-15  1.449e+00   0.000 1.000000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.366 on 20 degrees of freedom
## Multiple R-squared:  0.6706, Adjusted R-squared:  0.6212
## F-statistic: 13.57 on 3 and 20 DF,  p-value: 4.658e-05
```

```
predict(m1,
        newdata = data.frame(diet = "D"),
        interval = "prediction",
        level = 0.95)
```

```
##   fit      lwr      upr
## 1  61 55.76427 66.23573
```

The predicted coagulation time for a new animal assigned to Diet D is 61, with a 95% prediction interval of (55.76, 66.24).

```
library(lme4)
```

```
## Loading required package: Matrix
```

```
m2 <- lmer(coag ~ 1 + (1 | diet),
           data = coagulation, REML = TRUE)

summary(m2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: coag ~ 1 + (1 | diet)
##    Data: coagulation
##
## REML criterion at convergence: 115.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.18491 -0.59921  0.09332  0.54078  2.17508
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  diet     (Intercept) 11.692   3.419
```

```
##  Residual              5.599   2.366
## Number of obs: 24, groups:  diet, 4
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept)    64.01       1.78   35.96
```

```
predict(m2, newdata = data.frame(diet = "D"))
```

```
##        1
## 61.17017
```

```
pred <- predict(m2, newdata = data.frame(diet = "D"))
pred + c(-1, 1) * 1.96 * sigma(m2)
```

```
## [1] 56.53220 65.80813
```

Using a random effects model with REML, the predicted coagulation time for a new animal on Diet D is 61.17, with an approximate 95% prediction interval of (56.53, 65.81).

```
mu <- fixef(m2)[1]

sd_new <- sqrt(VarCorr(m2)$diet[1,1] + sigma(m2)^2)

mu + c(-1, 1) * 1.96 * sd_new
```

```
## [1] 55.86252 72.16281
```

The predicted coagulation time for a new animal on a new diet is 64.01, with an approximate 95% prediction interval of (55.86, 72.16). The interval is wider because it includes variation between diets and animals.

```
mu <- fixef(m2)[1]

sd_new <- sqrt(VarCorr(m2)$diet[1,1] + sigma(m2)^2)

mu
```

```
## (Intercept)
##    64.01266
```

```
mu + c(-1, 1) * 1.96 * sd_new
```

```
## [1] 55.86252 72.16281
```

The predicted coagulation time is 64.01, with an approximate 95% prediction interval of (55.86, 72.16). This is the same as part (d) because the model does not account for individual animal effects, and the previous diet has worn off.

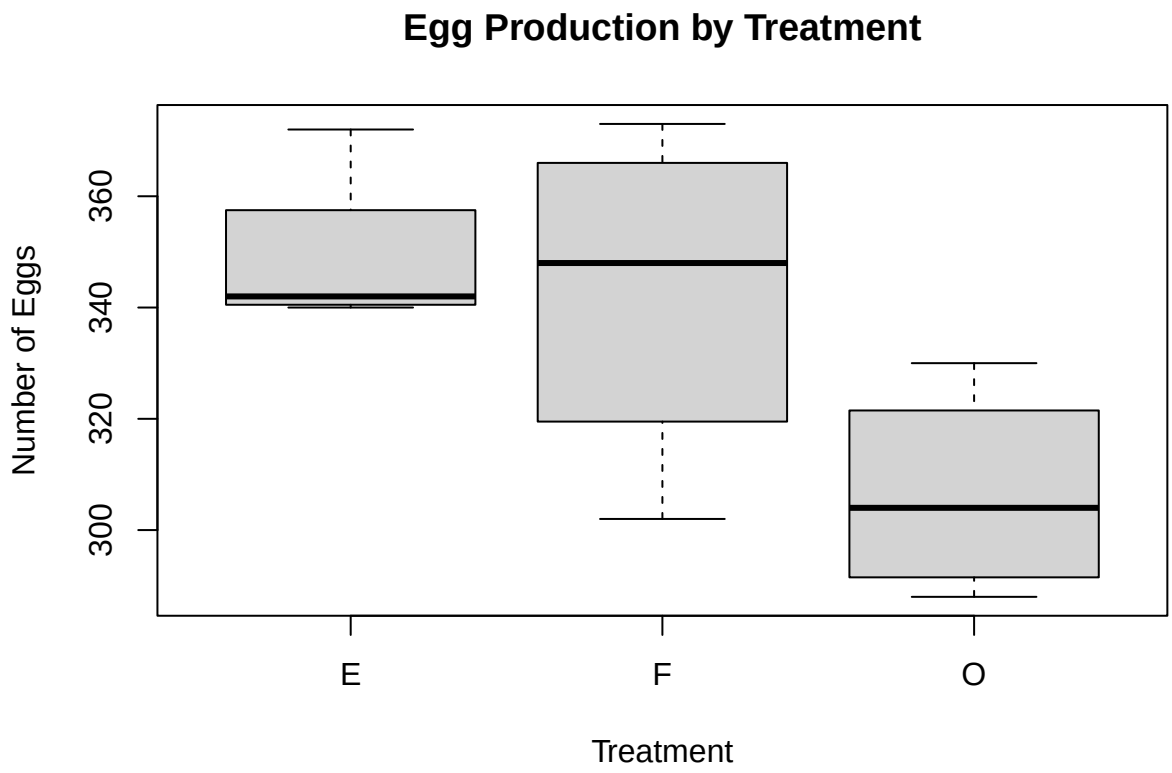
```r
data(eggprod)
head(eggprod)
```

```
##   treat block eggs
## 1     O     1  330
## 2     O     2  288
## 3     O     3  295
## 4     O     4  313
## 5     E     1  372
## 6     E     2  340
```

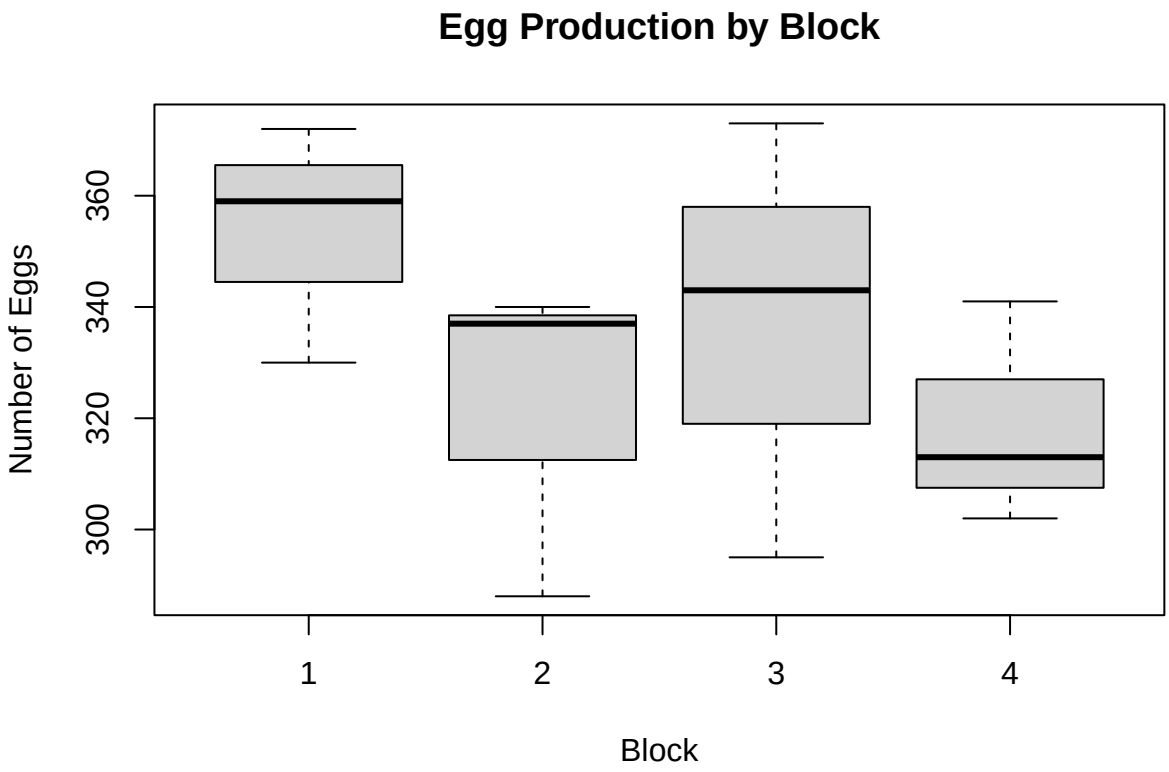
```r
str(eggprod)
```

```
## 'data.frame':    12 obs. of  3 variables:
##  $ treat: Factor w/ 3 levels "E","F","O": 3 3 3 3 1 1 1 1 2 2 ...
##  $ block: Factor w/ 4 levels "1","2","3","4": 1 2 3 4 1 2 3 4 1 2 ...
##  $ eggs : int  330 288 295 313 372 340 343 341 359 337 ...
```

```r
boxplot(eggs ~ treat, data = eggprod,
        main = "Egg Production by Treatment",
        xlab = "Treatment",
        ylab = "Number of Eggs")
```



```r
boxplot(eggs ~ block, data = eggprod,
        main = "Egg Production by Block",
        xlab = "Block",
        ylab = "Number of Eggs")
```



Treatments E and F generally produce more eggs than treatment O. Treatment F shows the most variability. Egg production also differs across blocks, with Block 1 having the highest production and Block 4 the lowest.

```r
m1 <- lm(eggs ~ treat + block, data = eggprod)

summary(m1)
```

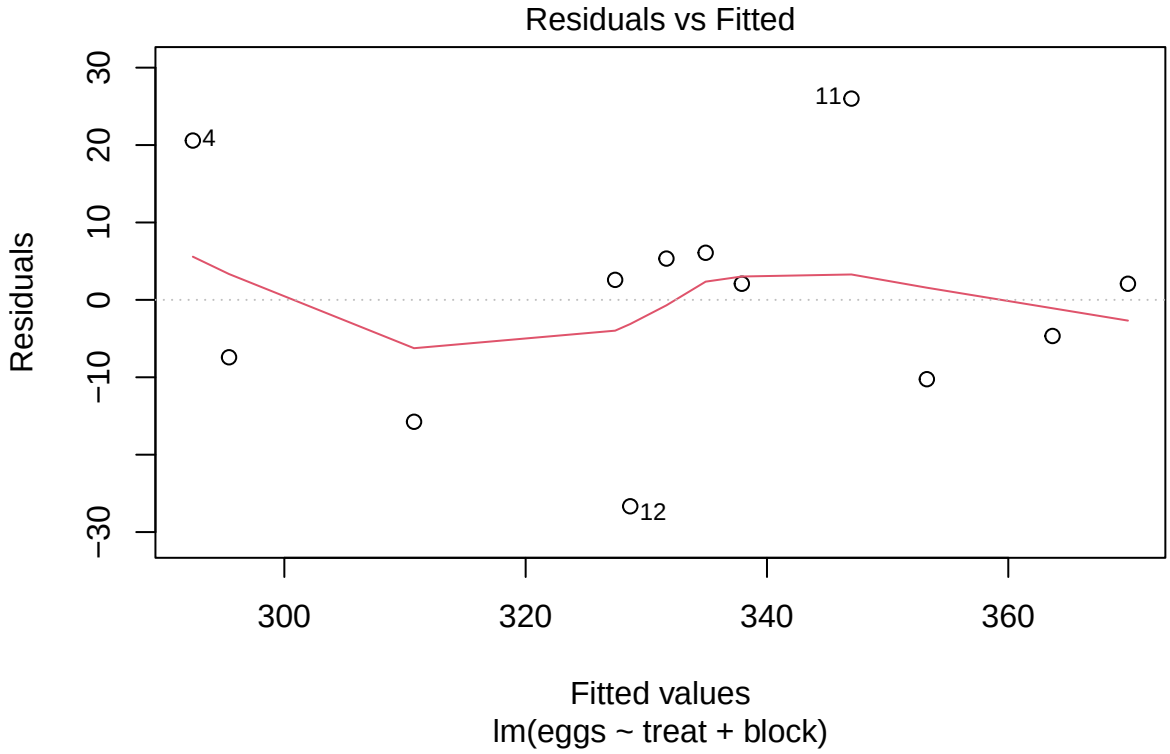
```
##
## Call:
## lm(formula = eggs ~ treat + block, data = eggprod)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -26.667  -8.125   2.083   5.521  26.000
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   369.92      13.91  26.596 1.87e-07 ***
## treatF         -6.25      13.91  -0.449   0.6690
## treatO        -42.50      13.91  -3.056   0.0224 *
```

```
## block2        -32.00      16.06  -1.992   0.0934 .
## block3        -16.67      16.06  -1.038   0.3394
## block4        -35.00      16.06  -2.179   0.0721 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.67 on 6 degrees of freedom
## Multiple R-squared:  0.7381, Adjusted R-squared:  0.5199
## F-statistic: 3.382 on 5 and 6 DF,  p-value: 0.08504
```

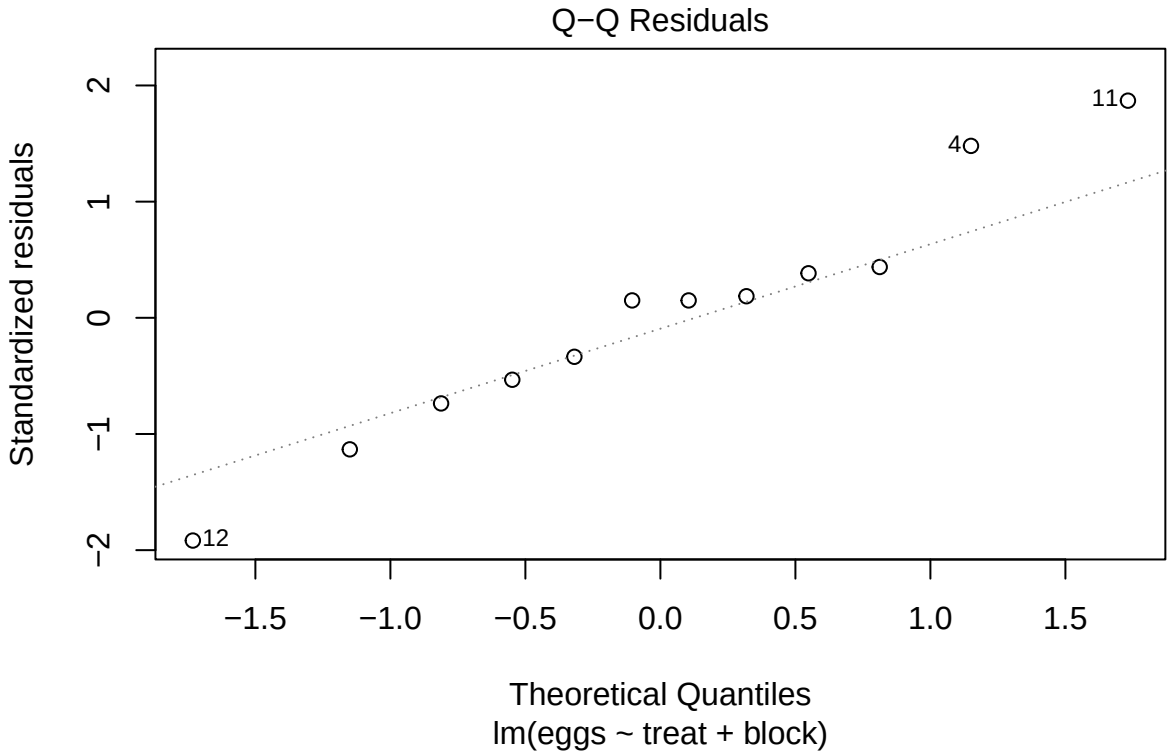
```
anova(m1)
```

```
## Analysis of Variance Table
##
## Response: eggs
##           Df Sum Sq Mean Sq F value  Pr(>F)
## treat      2 4212.5 2106.25  5.4437 0.04485 *
## block      3 2330.3  776.75  2.0075 0.21446
## Residuals  6 2321.5  386.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(m1, which = 1)
```



```r
plot(m1, which = 2)
```



The fixed effects model shows that treatment significantly affects egg production ($p = 0.0449$), while block does not ($p = 0.2145$). The diagnostic plots show some deviations, but no major violations of the model assumptions.

```r
library(lme4)

m2 <- lmer(eggs ~ treat + (1 | block),
           data = eggprod, REML = TRUE)

summary(m2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: eggs ~ treat + (1 | block)
##    Data: eggprod
##
## REML criterion at convergence: 85.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.71233 -0.47453 -0.02845  0.64196  1.42942
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  block    (Intercept) 129.9    11.40
```

```
##  Residual             386.9    19.67
## Number of obs: 12, groups:  block, 4
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept)   349.00      11.37  30.702
## treatF         -6.25      13.91  -0.449
## treatO        -42.50      13.91  -3.056
##
## Correlation of Fixed Effects:
##        (Intr) treatF
## treatF -0.612
## treatO -0.612  0.500
```

```
confint(m2, parm = c("treatF", "treatO"),
        method = "Wald")
```

```
##          2.5 %  97.5 %
## treatF -33.511  21.011
## treatO -69.761 -15.239
```

Treatment E has the highest predicted egg production (349 eggs). It is significantly better than treatment O, but not significantly different from treatment F. Therefore, we cannot conclude that E is better than both treatments.

```
library(pbkrtest)
full <- lmer(eggs ~ treat + (1 | block),
             data = eggprod, REML = FALSE)

reduced <- lmer(eggs ~ 1 + (1 | block),
                data = eggprod, REML = FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
KRmodcomp(full, reduced)
```

```
## large : eggs ~ treat + (1 | block)
## small : eggs ~ 1 + (1 | block)
##         stat    ndf    ddf F.scaling p.value
## Ftest 5.4437 2.0000 6.0000         1 0.04485 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Kenward-Roger F-test shows that treatment significantly affects egg production ($p = 0.04485$). This agrees with the fixed effects model from part (b), which gave the same p-value.

```
set.seed(123)
PBmodcomp(full, reduced, nsim = 1000)
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
## Bootstrap test; time: 13.20 sec; samples: 1000; extremes: 63;
## large : eggs ~ treat + (1 | block)
## eggs ~ 1 + (1 | block)
##          stat df p.value
## LRT    8.4245  2 0.01481 *
## PBtest 8.4245    0.06394 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The bootstrap test gave a p-value of 0.06394, which is not statistically significant at the 0.05 level. This differs from the Kenward-Roger test ($p = 0.04485$), which found a significant treatment effect.

```
library(RLRsim)
```

```
## Warning: package 'RLRsim' was built under R version 4.4.3
```

```
exactRLRT(m2)
```

```
##
##  simulated finite sample distribution of RLRT.
##
##  (p-value based on 10000 simulated values)
##
## data:
## RLRT = 0.51536, p-value = 0.2138
```

The random effects test gave a p-value of 0.2138, showing that block is not statistically significant. This agrees with the fixed effects model from part (b), which gave a p-value of 0.2145.